Lab Midterm – Web Technologies (CSC336)

Name: Muhammad Hammad Sarwar **Registration No.:** FA24-BCS-068

N	BP = 600 + N	H = (N×7) mod 360	K = (N mod 3) + 3	Cart limit = (N mod 4) + 2	Delivery dates
68	668px	116 → hsl(116, 70%, 45%)	5	2	2026-10-29 to 2027-01-05

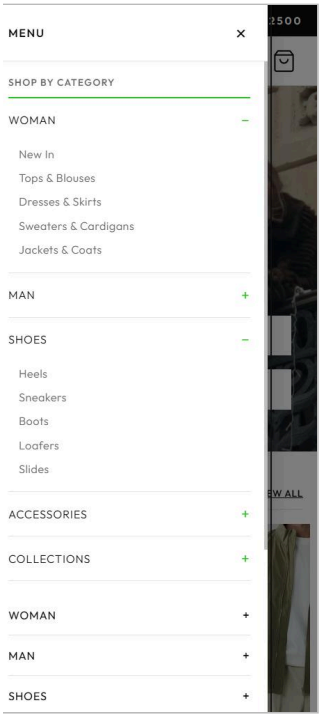
Project: Lama Retail replica (Assignment 2) copied to `midterm/` (commit "midterm: baseline copy"). Bootstrap 5.3.3 kept. Screenshots at 375px, BP = 668px and 1280px. Task 1 and 2 use no JavaScript of my own; Task 3 is in `script.js`.

Contents

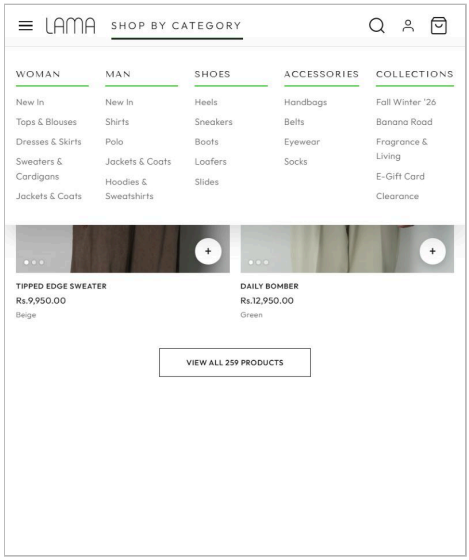
1. Screenshots – Task 1 (mega menu), Task 2 (checkout validation), Task 3 (cart badge + mini cart)
2. Code – Task 1, Task 2, Task 3 (as text)

Task 1 – Mega menu – screenshots

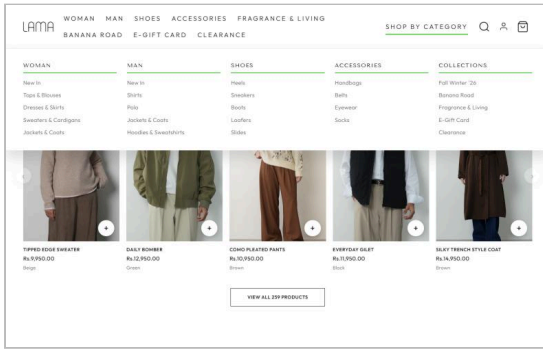
1280px and 668px: “Shop by Category” open (full-width panel, K = 5 columns, above the product grid). 375px: the same groups as a <details> accordion in the Bootstrap offcanvas menu (Woman and Shoes opened, the rest closed).



375px – accordion in hamburger menu



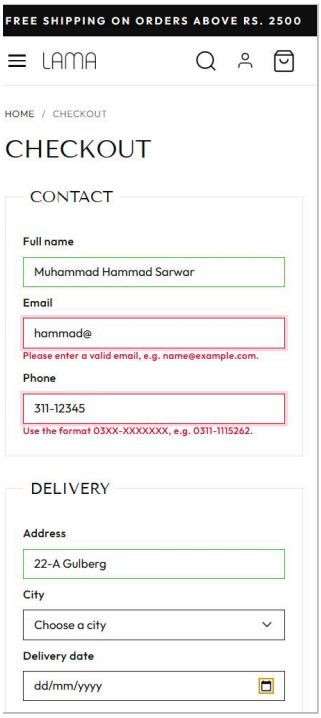
668px (BP) – panel open



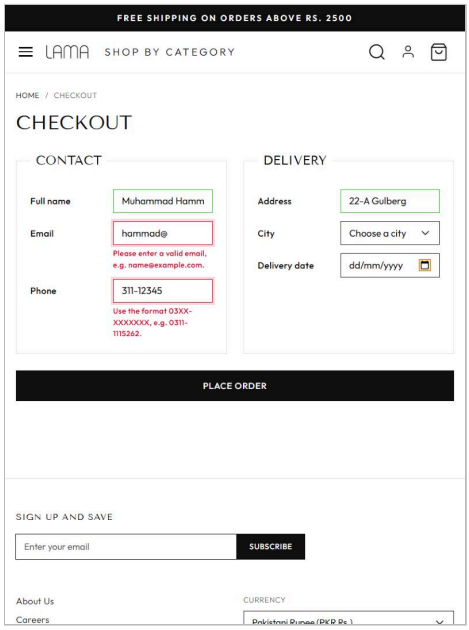
1280px – panel open

Task 2 – Checkout form – screenshots

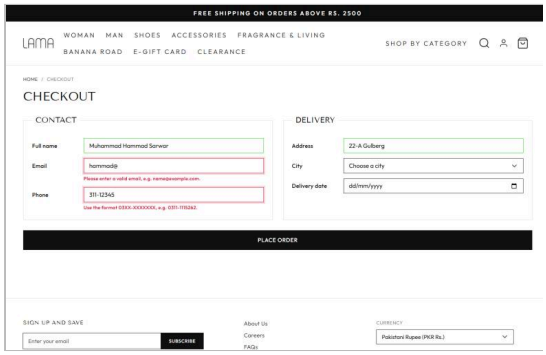
After typing an invalid email and phone: red border + message only on the touched invalid fields; valid fields get the green accent border; untouched fields stay neutral. 375px: one column, labels above inputs. 668px and 1280px: fieldsets side by side, labels in a left column. Submit button full width.



375px – one column



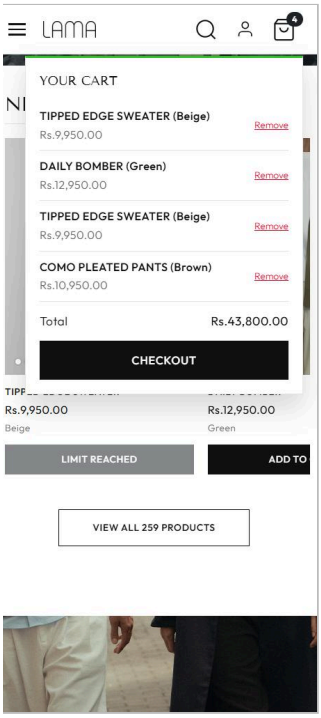
668px (BP) – side by side



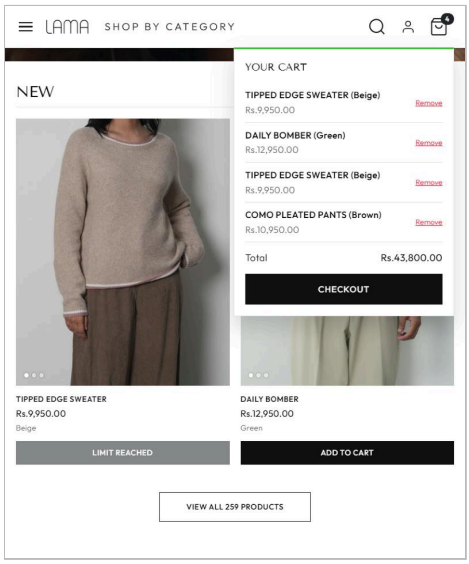
1280px – side by side

Task 3 – Cart badge and mini cart – screenshots

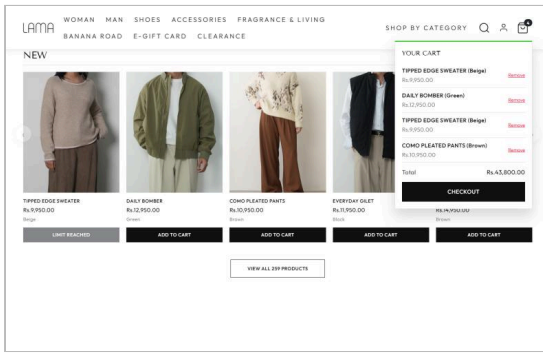
After using the cart: Tipped Edge Sweater added twice (button now “Limit reached”, disabled), Daily Bomber and Como Pleated Pants added once. Badge shows 4; mini cart lists each item with its price and Remove, total Rs.43,800.00 calculated with reduce().



375px – mini cart open



668px (BP) – mini cart open



1280px – mini cart open

Task 1 – code added / changed

index.html / product.html – "Shop by Category" item inside the existing <nav>

```
                    <!-- MIDTERM TASK 1: "Shop by Category" mega menu - HTML + CSS only (css/mega-menu.css).
                         Opens on :hover or :focus-within (Tab), from BP = 668px and wider. -->
                    <ul class="navbar-nav shop-nav">
                        <li class="nav-item shop-menu">
                            <a class="nav-link shop-menu__link" href="#shop-panel" aria-haspopup="true">Shop by Category</a>
                            <div class="shop-panel" id="shop-panel">
                                <div class="container shop-panel__grid">
                                    <div class="shop-panel__col">
                                        <h2 class="shop-panel__heading" id="shop-heading-woman">Woman</h2>
                                        <ul class="shop-panel__list" aria-labelledby="shop-heading-woman">
                                            <li><a href="index.html#section-new">New In</a></li>
                                            <li><a href="#">Tops &amp;amp; Blouses</a></li>
                                            <li><a href="#">Dresses &amp;amp; Skirts</a></li>
                                            <li><a href="index.html#section-sweaters">Sweaters &amp;amp; Cardigans</a></li>
                                            <li><a href="index.html#section-jackets">Jackets &amp;amp; Coats</a></li>
                                        </ul>
                                    </div>
                                    <div class="shop-panel__col">
                                        <h2 class="shop-panel__heading" id="shop-heading-man">Man</h2>
                                        <ul class="shop-panel__list" aria-labelledby="shop-heading-man">
                                            <li><a href="index.html#section-new">New In</a></li>
                                            <li><a href="#">Shirts</a></li>
                                            <li><a href="#">Polo</a></li>
                                            <li><a href="index.html#section-jackets">Jackets &amp;amp; Coats</a></li>
                                            <li><a href="#">Hoodies &amp;amp; Sweatshirts</a></li>
                                        </ul>
                                    </div>
                                    <div class="shop-panel__col">
                                        <h2 class="shop-panel__heading" id="shop-heading-shoes">Shoes</h2>
                                        <ul class="shop-panel__list" aria-labelledby="shop-heading-shoes">
                                            <li><a href="index.html#section-shoes">Heels</a></li>
                                            <li><a href="index.html#section-shoes">Sneakers</a></li>
                                            <li><a href="index.html#section-shoes">Boots</a></li>
                                            <li><a href="index.html#section-shoes">Loafers</a></li>
                                            <li><a href="#">Slides</a></li>
                                        </ul>
                                    </div>
                                    <div class="shop-panel__col">
                                        <h2 class="shop-panel__heading" id="shop-heading-accessories">Accessories</h2>
                                        <ul class="shop-panel__list" aria-labelledby="shop-heading-accessories">
                                            <li><a href="index.html#section-bags">Handbags</a></li>
                                            <li><a href="#">Belts</a></li>
                                            <li><a href="#">Eyewear</a></li>
                                            <li><a href="#">Socks</a></li>
                                        </ul>
                                    </div>
                                    <div class="shop-panel__col">
                                        <h2 class="shop-panel__heading" id="shop-heading-collections">Collections</h2>
                                        <ul class="shop-panel__list" aria-labelledby="shop-heading-collections">
                                            <li><a href="#">Fall Winter '26</a></li>
                                            <li><a href="#">Banana Road</a></li>
                                            <li><a href="#">Fragrance &amp;amp; Living</a></li>
                                            <li><a href="#">E-Gift Card</a></li>
                                            <li><a href="#">Clearance</a></li>
                                        </ul>
                                    </div>
                                </div>
                            </div>
                        </li>
                    </ul>
```

index.html / product.html – hamburger now opened by Bootstrap's own offcanvas script

```
                    <button type="button" id="mobileMenuBtn" class="mobile-menu-btn" aria-label="Open mobile navigation menu"
data-bs-toggle="offcanvas" data-bs-target="#mobileDrawer" aria-controls="mobileDrawer">
...
    <aside id="mobileDrawer" class="offcanvas offcanvas-start mobile-drawer" tabindex="-1" aria-label="Navigation Menu">
...
            <button type="button" id="mobileDrawerClose" class="mobile-drawer-close" data-bs-dismiss="offcanvas" aria-
label="Close menu">&times;</button>
...
    <script src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.3/dist/js/bootstrap.bundle.min.js"
```

```
            integrity="sha384-YvpcrYf0tY3lHB60NNkmXc5s9fDVZLESaAA55NDzOxhy9GkcIdslK1eN7N6jIeHz" crossorigin="anonymous">
</script>
```

index.html / product.html – the same K = 5 groups as a <details> accordion (below BP)

```
            <!-- MIDTERM TASK 1: the same K = 5 groups as a <details> accordion (below BP = 668px only).
                 Closed by default; each <details> opens on its own. No JavaScript. -->
            <nav class="shop-accordion" aria-label="Shop by Category">
                <p class="shop-accordion__title">Shop by Category</p>
                <details class="shop-accordion__group">
                    <summary>Woman</summary>
                    <ul>
                        <li><a href="index.html#section-new">New In</a></li>
                        <li><a href="#">Tops &amp; Blouses</a></li>
                        <li><a href="#">Dresses &amp; Skirts</a></li>
                        <li><a href="index.html#section-sweaters">Sweaters &amp; Cardigans</a></li>
                        <li><a href="index.html#section-jackets">Jackets &amp; Coats</a></li>
                    </ul>
                </details>
                <details class="shop-accordion__group">
                    <summary>Man</summary>
                    <ul>
                        <li><a href="index.html#section-new">New In</a></li>
                        <li><a href="#">Shirts</a></li>
                        <li><a href="#">Polo</a></li>
                        <li><a href="index.html#section-jackets">Jackets &amp; Coats</a></li>
                        <li><a href="#">Hoodies &amp; Sweatshirts</a></li>
                    </ul>
                </details>
                <details class="shop-accordion__group">
                    <summary>Shoes</summary>
                    <ul>
                        <li><a href="index.html#section-shoes">Heels</a></li>
                        <li><a href="index.html#section-shoes">Sneakers</a></li>
                        <li><a href="index.html#section-shoes">Boots</a></li>
                        <li><a href="index.html#section-shoes">Loafers</a></li>
                        <li><a href="#">Slides</a></li>
                    </ul>
                </details>
                <details class="shop-accordion__group">
                    <summary>Accessories</summary>
                    <ul>
                        <li><a href="index.html#section-bags">Handbags</a></li>
                        <li><a href="#">Belts</a></li>
                        <li><a href="#">Eyewear</a></li>
                        <li><a href="#">Socks</a></li>
                    </ul>
                </details>
                <details class="shop-accordion__group">
                    <summary>Collections</summary>
                    <ul>
                        <li><a href="#">Fall Winter '26</a></li>
                        <li><a href="#">Banana Road</a></li>
                        <li><a href="#">Fragrance &amp; Living</a></li>
                        <li><a href="#">E-Gift Card</a></li>
                        <li><a href="#">Clearance</a></li>
                    </ul>
                </details>
            </nav>
```

css/mega-menu.css (new file, complete)

```
/* ============================================================================
   css/mega-menu.css - MIDTERM TASK 1: "Shop by Category" mega menu (HTML + CSS only)

   My numbers (roll no. FA24-BCS-068):  N = 68
     BP = 600 + N = 668px      -> the panel works from 668px; below it the <details> accordion is used
     K  = (N mod 3) + 3 = 5    -> 5 columns in the panel / 5 accordion groups
     H  = (N x 7) mod 360 = 116 -> accent colour hsl(116, 70%, 45%)

   To change BP: search this file for "BP" (CSS variables can't be used inside @media).
   To change K:  edit grid-template-columns in .shop-panel__grid.
   ============================================================================ */

:root {
    --accent: hsl(116, 70%, 45%);   /* H = 116 */
}
```

```
/* 1. The desktop <nav> now shows from BP (it used to show only from 1024px).
      Between BP and 1023px only "Shop by Category" is visible; the other 8 links stay in the hamburger. */
@media (min-width: 668px) {                    /* BP */
    .desktop-nav {
        display: flex;
        align-items: center;
        flex: 1;
        gap: var(--space-6);
        margin: 0 var(--space-4) 0 var(--space-6);
    }
}

@media (max-width: 1023.98px) {
    .desktop-nav .nav-list {
        display: none;                         /* the JS-built links still appear only from 1024px */
    }
}

/* 2. Position: the panel is placed relative to the sticky header (full page width),
      so the navbar and the menu item must NOT be the positioned parent. */
.header-inner.navbar {
    position: static;                          /* Bootstrap's .navbar is position: relative */
}

.shop-menu.nav-item {
    position: static;                          /* our .nav-item is position: relative */
}

/* 3. The panel: hidden by default, full width under the navbar, above everything */
.shop-panel {
    position: absolute;                        /* out of the page flow -> does not push content down */
    top: 100%;                                 /* starts at the bottom edge of the header */
    left: 0;
    right: 0;                                  /* left 0 + right 0 = full width of the header */
    z-index: 10;                               /* inside the header, which is z-index 500 above the page */
    background-color: var(--color-white);
    border-top: 1px solid var(--color-border);
    box-shadow: 0 18px 30px rgba(0, 0, 0, 0.12);
    padding: var(--space-8) 0;

    visibility: hidden;                        /* hidden = not clickable and not reachable with Tab */
    opacity: 0;
    /* When CLOSING, wait 0.25s before hiding: the pointer has time to travel from
       "Shop by Category" across the gap into the panel without it closing. */
    transition: opacity 0.15s ease, visibility 0s linear 0.25s;
}

/* OPEN: mouse over the item or the panel (:hover), or keyboard focus anywhere inside it (:focus-within).
   The panel is inside the <li>, so hovering the panel still counts as hovering the <li>. */
.shop-menu:hover > .shop-panel,
.shop-menu:focus-within > .shop-panel {
    visibility: visible;
    opacity: 1;
    transition-delay: 0s;                      /* open immediately */
}

/* Below BP the panel never shows (the accordion in the hamburger menu is used instead) */
@media (max-width: 667.98px) {                 /* BP - 0.02 */
    .shop-nav {
        display: none;
    }
}

/* 4. K columns, each a list with a heading */
.shop-panel__grid {
    display: grid;
    grid-template-columns: repeat(5, 1fr);     /* K = 5 */
    gap: var(--space-6);
}

.shop-panel__heading {
    font-size: var(--font-size-sm);
    text-transform: uppercase;
    letter-spacing: var(--letter-spacing-wider);
    padding-bottom: var(--space-2);
    margin-bottom: var(--space-3);
    border-bottom: 2px solid var(--accent);    /* accent colour from H */
}

.shop-panel__list li + li {
```

```
    margin-top: var(--space-2);
}

.shop-panel__list a {
    font-size: var(--font-size-sm);
    color: var(--color-text-muted);
}

.shop-panel__list a:hover,
.shop-panel__list a:focus-visible {
    color: var(--color-text-main);
    text-decoration: underline;
    text-decoration-color: var(--accent);
}

.shop-menu:hover > .shop-menu__link,
.shop-menu:focus-within > .shop-menu__link {
    box-shadow: inset 0 -2px 0 var(--accent);   /* accent underline while the panel is open */
}

/* 5. Below BP: the same K groups as a <details> accordion inside the hamburger menu */
.shop-accordion {
    margin-bottom: var(--space-4);
}

@media (min-width: 668px) {                     /* BP */
    .shop-accordion {
        display: none;                          /* from BP the desktop panel is used instead */
    }
}

.shop-accordion__title {
    font-size: var(--font-size-xs);
    font-weight: 500;
    text-transform: uppercase;
    letter-spacing: var(--letter-spacing-wider);
    color: var(--color-text-muted);
    padding-bottom: var(--space-2);
    border-bottom: 2px solid var(--accent);
}

.shop-accordion__group {
    border-bottom: 1px solid var(--color-border);
}

.shop-accordion__group summary {
    display: flex;
    justify-content: space-between;
    align-items: center;
    padding: var(--space-3) 0;
    font-size: var(--font-size-sm);
    text-transform: uppercase;
    letter-spacing: var(--letter-spacing-wide);
    cursor: pointer;
    list-style: none;                           /* hide the default triangle (Firefox) */
}

.shop-accordion__group summary::-webkit-details-marker {
    display: none;                              /* hide the default triangle (Chrome / Safari) */
}

.shop-accordion__group summary::after {
    content: '+';
    color: var(--accent);
    font-size: 1.1rem;
}

.shop-accordion__group[open] summary::after {
    content: '\2212';                           /* minus sign when open */
}

.shop-accordion__group ul {
    padding: 0 0 var(--space-3) var(--space-3);
}

.shop-accordion__group li a {
    display: block;
    padding: var(--space-1) 0;
    font-size: var(--font-size-sm);
    color: var(--color-text-muted);
```

```
}

/* 6. Phone menu = Bootstrap offcanvas now: let Bootstrap's z-index and slide animation apply
      (our old .mobile-drawer rule set a lower z-index meant for the removed custom backdrop) */
.mobile-drawer.offcanvas {
    z-index: var(--bs-offcanvas-zindex);
    transition: transform 0.3s ease-in-out;
}
```

js/menu.js – our own hamburger code removed (Bootstrap does it now); replaced by this comment

```
        // 3. Mobile Drawer: opened and closed by Bootstrap's own offcanvas script (data-bs-toggle="offcanvas"
        //    on the hamburger button in the HTML), which also handles Esc, the backdrop and focus. (Midterm Task 1)
```

Task 2 – code added / changed

index.html / product.html – cart icon is now a link to checkout.html

```
                    <!-- MIDTERM TASK 2: the cart icon is a real link to the checkout page -->
                    <a href="checkout.html" id="cartTriggerBtn" class="header-icon-btn" aria-label="View cart and checkout">
```

checkout.html – <main> with the form (header and footer are the same as the other pages)

```
         MIDTERM TASK 2: CHECKOUT FORM - HTML5 validation + CSS only (css/checkout.css), no JavaScript
         ========================================================================== -->
    <main id="MainContent" role="main" class="checkout-page">
        <div class="container">

            <nav aria-label="Breadcrumb">
                <ol class="breadcrumb checkout-breadcrumb">
                    <li class="breadcrumb-item"><a href="index.html">Home</a></li>
                    <li class="breadcrumb-item active" aria-current="page">Checkout</li>
                </ol>
            </nav>

            <h1 class="checkout-title">Checkout</h1>

            <!-- Shown only after a valid submit: the form goes to #order-confirmed and CSS :target shows this box -->
            <p id="order-confirmed" class="order-confirmed" role="status">
                Thank you! Your order details were accepted. (Demo form &amp;ndash; nothing is sent anywhere.)
            </p>

            <!-- The browser checks required / type / pattern / minlength / min / max before it lets the form submit.
                 Inputs have no name attribute, so no personal data is added to the URL. -->
            <form class="checkout-form" action="#order-confirmed">

                <fieldset class="checkout-fieldset">
                    <legend>Contact</legend>
                        <label for="full-name">Full name</label>
                        <input id="full-name" class="form-control" type="text" required minlength="3" autocomplete="name"
placeholder="e.g. Muhammad Hammad Sarwar" aria-describedby="full-name-error">
                        <span class="field-error" id="full-name-error">Please enter your full name (at least 3 letters).</span>
                        <label for="email">Email</label>
                        <input id="email" class="form-control" type="email" required autocomplete="email"
placeholder="name@example.com" aria-describedby="email-error">
                        <span class="field-error" id="email-error">Please enter a valid email, e.g. name@example.com.</span>
                        <label for="phone">Phone</label>
                        <input id="phone" class="form-control" type="tel" required pattern="03[0-9]{2}-[0-9]{7}"
placeholder="03XX-XXXXXXX" title="Format: 03XX-XXXXXXX" autocomplete="tel" aria-describedby="phone-error">
                        <span class="field-error" id="phone-error">Use the format 03XX-XXXXXXX, e.g. 0311-1115262.</span>
                </fieldset>

                <fieldset class="checkout-fieldset">
                    <legend>Delivery</legend>
                        <label for="address">Address</label>
                        <input id="address" class="form-control" type="text" required minlength="5" autocomplete="street-
address" placeholder="House, street, area" aria-describedby="address-error">
                        <span class="field-error" id="address-error">Please enter your street address.</span>
                        <label for="city">City</label>
                        <select id="city" class="form-select" required aria-describedby="city-error">
                            <option value="">Choose a city</option>
                            <option>Lahore</option>
                            <option>Karachi</option>
                            <option>Islamabad</option>
                            <option>Rawalpindi</option>
                            <option>Faisalabad</option>
                            <option>Multan</option>
                            <option>Peshawar</option>
                        </select>
                        <span class="field-error" id="city-error">Please choose a city.</span>
                        <label for="delivery-date">Delivery date</label>
                        <input id="delivery-date" class="form-control" type="date" required min="2026-10-29" max="2027-01-05"
aria-describedby="delivery-date-error">
                        <span class="field-error" id="delivery-date-error">Choose a date from 29 Oct 2026 to 5 Jan 2027.</span>
                </fieldset>

                <button type="submit" class="btn btn-dark checkout-submit">Place order</button>
            </form>
```

```
        </div>
    </main>
```

css/checkout.css (new file, complete)

```
/* ========================================================================
   css/checkout.css - MIDTERM TASK 2: checkout form (HTML5 validation + CSS only, no JavaScript)

   BP = 668px (search "BP" to change it)
     below BP : one column, labels above inputs
     from BP  : the two fieldsets side by side; inside each, labels in a left column (CSS Grid)
   ======================================================================== */

.checkout-page {
    padding: var(--space-6) 0 var(--space-12);
}

.checkout-breadcrumb {
    font-size: var(--font-size-xs);
    text-transform: uppercase;
    letter-spacing: var(--letter-spacing-wide);
    margin-bottom: var(--space-4);
}

.checkout-title {
    font-size: var(--font-size-2xl);
    text-transform: uppercase;
    margin-bottom: var(--space-6);
}

/* 1. Page layout: the form is a grid - 1 column on phones, 2 columns (Contact | Delivery) from BP */
.checkout-form {
    display: grid;
    grid-template-columns: 1fr;
    gap: var(--space-6);
}

@media (min-width: 668px) {                     /* BP */
    .checkout-form {
        grid-template-columns: 1fr 1fr;         /* the two fieldsets sit side by side */
    }
}

/* Submit button: full width at every size, under both fieldsets */
.checkout-submit {
    grid-column: 1 / -1;                        /* span all columns of the form grid */
    width: 100%;
}

/* 2. Each fieldset is its own grid: label | input (+ error message under the input) */
.checkout-fieldset {
    display: grid;
    grid-template-columns: 1fr;                 /* below BP: label above input */
    row-gap: var(--space-2);
    align-items: center;
    align-content: start;
    border: 1px solid var(--color-border);
    padding: var(--space-5);
    margin: 0;
}

@media (min-width: 668px) {                     /* BP */
    .checkout-fieldset {
        grid-template-columns: 6.5rem minmax(0, 1fr);   /* labels in a column on the left of their inputs */
        column-gap: var(--space-4);
        row-gap: var(--space-3);
    }

    .checkout-fieldset .field-error {
        grid-column: 2;                         /* message sits under its input, not under the label */
    }
}

.checkout-fieldset legend {
    float: none;
    width: auto;
    padding: 0 var(--space-2);
    margin-bottom: var(--space-2);
    font-family: var(--font-heading);
```

```
    font-size: var(--font-size-lg);
    text-transform: uppercase;
}

.checkout-fieldset label {
    font-size: var(--font-size-sm);
    font-weight: 500;
}

.checkout-fieldset .form-control,
.checkout-fieldset .form-select {
    font-size: var(--font-size-base);
    border-color: var(--color-border-dark);
}

.checkout-fieldset .form-control:focus,
.checkout-fieldset .form-select:focus {
    border-color: var(--accent);               /* accent colour hsl(116, 70%, 45%) from css/mega-menu.css */
    box-shadow: 0 0 0 0.2rem hsl(116 70% 45% / 0.25);
}

/* 3. Validation messages: hidden until the user has interacted with the field */
.field-error {
    display: none;
    color: var(--color-sale);
    font-size: var(--font-size-xs);
    font-weight: 500;
    margin-top: calc(var(--space-2) * -1);
}

/* :user-invalid = invalid AND the user has already typed in / left the field (not on page load) */
.checkout-form .form-control:user-invalid,
.checkout-form .form-select:user-invalid {
    border-color: var(--color-sale);           /* red border */
    box-shadow: 0 0 0 0.2rem rgba(217, 4, 41, 0.15);
}

.checkout-form .form-control:user-invalid + .field-error,
.checkout-form .form-select:user-invalid + .field-error {
    display: block;                            /* "+" = the message right after the invalid field */
}

/* Valid fields the user filled in get a green (accent) border */
.checkout-form .form-control:user-valid,
.checkout-form .form-select:user-valid {
    border-color: var(--accent);
}

/* Fallback for older browsers without :user-invalid:
   :placeholder-shown is true while the field is empty, so this only fires after typing */
@supports not selector(:user-invalid) {
    .checkout-form .form-control:not(:placeholder-shown):invalid {
        border-color: var(--color-sale);
    }

    .checkout-form .form-control:not(:placeholder-shown):invalid + .field-error {
        display: block;
    }
}

/* 4. Confirmation box: shown only when the URL ends with #order-confirmed (after a valid submit) */
.order-confirmed {
    display: none;
    padding: var(--space-4);
    margin-bottom: var(--space-6);
    border: 1px solid var(--accent);
    border-left-width: 4px;
    background-color: hsl(116 70% 45% / 0.08);
    font-weight: 500;
}

.order-confirmed:target {
    display: block;
}
```

Task 3 – code added / changed

script.js (new file, complete)

```
/**
 * script.js - MIDTERM TASK 3: cart badge + mini cart (plain JavaScript, no libraries)
 *
 * N = 68  ->  each product may be added at most (N mod 4) + 2 = (68 mod 4) + 2 = 0 + 2 = 2 times
 */
(function () {
    "use strict";

    const MAX_PER_PRODUCT = (68 % 4) + 2;   // = 2

    // The cart: an array of { key, name, colour, price } objects (one entry per item added)
    let cart = [];

    // "Rs.9,950.00" -> 9950   (remove the "Rs." prefix and the thousands commas)
    function parsePrice(text) {
        return Number(text.replace("Rs.", "").replace(/,/g, "").trim());
    }

    // 9950 -> "Rs.9,950.00"
    function formatPrice(n) {
        return "Rs." + n.toLocaleString("en-PK", { minimumFractionDigits: 2, maximumFractionDigits: 2 });
    }

    // Same product = same name AND colour (e.g. Short Knit Polo comes in Brown and Light Blue)
    function makeKey(name, colour) {
        return name + "|" + colour;
    }

    // How many of this product are in the cart?
    function countOf(key) {
        return cart.filter(item => item.key === key).length;
    }

    document.addEventListener("DOMContentLoaded", function () {
        const cartIcon = document.getElementById("cartTriggerBtn");
        const badge = document.getElementById("headerCartBadge");
        const panel = document.getElementById("miniCart");
        const list = document.getElementById("miniCartList");
        const emptyText = document.getElementById("miniCartEmpty");
        const totalEl = document.getElementById("miniCartTotal");
        if (!cartIcon || !badge || !panel) return;

        cartIcon.setAttribute("aria-controls", "miniCart");
        cartIcon.setAttribute("aria-expanded", "false");

        // ---------- Add an item (respecting the limit), then redraw everything
        function addItem(name, colour, price) {
            const key = makeKey(name, colour);
            if (countOf(key) >= MAX_PER_PRODUCT) return false;
            cart.push({ key: key, name: name, colour: colour, price: price });
            render();
            return true;
        }

        // ---------- Remove one item by its position in the array
        function removeItem(index) {
            cart.splice(index, 1);
            render();
        }

        // ---------- Redraw: badge, mini-cart list, total, and every Add to Cart button
        function render() {
            // Badge = number of items in the cart (CSS hides it when data-count is "0")
            badge.textContent = cart.length;
            badge.setAttribute("data-count", cart.length);
            cartIcon.setAttribute("aria-label", `View cart, ${cart.length} item${cart.length === 1 ? "" : "s"}`);

            // List
            list.innerHTML = "";
            cart.forEach(function (item, index) {
                const li = document.createElement("li");
                li.className = "mini-cart__item";

                const info = document.createElement("span");
```

```
        info.className = "mini-cart__name";
        info.textContent = `${item.name} (${item.colour})`;

        const price = document.createElement("span");
        price.className = "mini-cart__price";
        price.textContent = formatPrice(item.price);

        const removeBtn = document.createElement("button");
        removeBtn.type = "button";
        removeBtn.className = "mini-cart__remove";
        removeBtn.textContent = "Remove";
        removeBtn.setAttribute("aria-label", `Remove ${item.name} (${item.colour})`);
        removeBtn.addEventListener("click", function (e) {
            e.stopPropagation();      // keep the panel open
            removeItem(index);
        });

        li.append(info, price, removeBtn);
        list.appendChild(li);
    });
    emptyText.hidden = cart.length > 0;

    // Total with reduce(): start at 0 and add each item's price
    const total = cart.reduce((sum, item) => sum + item.price, 0);
    totalEl.textContent = formatPrice(total);

    // Buttons: "Limit reached" + disabled once a product is in the cart MAX_PER_PRODUCT times
    document.querySelectorAll(".btn-add-to-cart").forEach(function (btn) {
        const card = btn.closest(".product-card");
        const key = makeKey(cardName(card), cardColour(card));
        const full = countOf(key) >= MAX_PER_PRODUCT;
        btn.disabled = full;
        btn.textContent = full ? "Limit reached" : "Add to Cart";
    });
}

// Read the product details from the card itself
function cardName(card) {
    return card.querySelector(".product-title").textContent.trim();
}

function cardColour(card) {
    const el = card.querySelector(".product-color-label");
    return el ? el.textContent.trim() : "";
}

// ---------- Click on any "Add to Cart" button (event delegation: one listener for all cards,
//            including cards added later)
document.addEventListener("click", function (e) {
    const btn = e.target.closest(".btn-add-to-cart");
    if (!btn || btn.disabled) return;
    const card = btn.closest(".product-card");
    const price = parsePrice(card.querySelector(".product-price").textContent);
    addItem(cardName(card), cardColour(card), price);
});

// ---------- Product page: its own Add to Cart form sends a "cart:add" event (js/product-page.js)
window.addEventListener("cart:add", function (e) {
    const d = e.detail || {};
    const qty = Number.isInteger(d.quantity) ? d.quantity : 1;
    for (let i = 0; i < qty; i++) {
        if (!addItem(d.name, d.color, d.price)) break;   // stop at the limit
    }
    openPanel();
});

// ---------- Mini-cart panel: the cart icon toggles it (instead of going to checkout.html)
function openPanel() {
    panel.hidden = false;
    cartIcon.setAttribute("aria-expanded", "true");
}

function closePanel() {
    panel.hidden = true;
    cartIcon.setAttribute("aria-expanded", "false");
}

cartIcon.addEventListener("click", function (e) {
    e.preventDefault();               // the icon is a link to checkout.html (Task 2); JS shows the mini cart instead
    if (panel.hidden) {
```

```
                openPanel();
            } else {
                closePanel();
            }
        });

        // Close with Esc or a click outside the panel
        document.addEventListener("keydown", function (e) {
            if (e.key === "Escape" && !panel.hidden) closePanel();
        });

        document.addEventListener("click", function (e) {
            if (!panel.hidden && !panel.contains(e.target) && !cartIcon.contains(e.target)) closePanel();
        });

        render();
    });
})();
```

index.html / product.html – mini-cart panel under the cart icon

```
                    <!-- MIDTERM TASK 3: mini cart panel under the cart icon (filled and opened by script.js) -->
                    <div id="miniCart" class="mini-cart" role="region" aria-label="Mini cart" hidden>
                        <p class="mini-cart__title">Your cart</p>
                        <p id="miniCartEmpty" class="mini-cart__empty">Your cart is empty.</p>
                        <ul id="miniCartList" class="mini-cart__list"></ul>
                        <p class="mini-cart__total"><span>Total</span> <strong id="miniCartTotal">Rs.0.00</strong></p>
                        <a href="checkout.html" class="btn btn-dark w-100 mini-cart__checkout">Checkout</a>
                    </div>
```

index.html / product.html – script.js linked (old cart.js / quick-add.js removed)

```
    <script src="script.js"></script>                  <!-- Midterm Task 3: cart badge + mini cart -->
```

js/products.js – Add to Cart button added to every product card (the "+" quick-add button removed)

```
        // "Add to Cart" button (midterm Task 3): script.js listens for clicks on .btn-add-to-cart
        // and reads the product name and price from this card
        const addBtn = document.createElement("button");
        addBtn.type = "button";
        addBtn.className = "btn btn-dark btn-sm w-100 btn-add-to-cart";
        addBtn.textContent = "Add to Cart";
        meta.appendChild(addBtn);

        card.appendChild(meta);
```

css/mini-cart.css (new file, complete)

```
/* ========================================================================
   css/mini-cart.css - MIDTERM TASK 3: "Add to Cart" buttons on cards + mini-cart panel (script.js)
   ======================================================================== */

/* Add to Cart button at the bottom of every product card */
.btn-add-to-cart {
    margin-top: var(--space-2);
    padding: 8px 12px;
    font-size: var(--font-size-xs);
}

.btn-add-to-cart:disabled {
    opacity: 0.55;                      /* "Limit reached" */
}

/* The panel hangs under the cart icon (the icon group is the positioned parent) */
.header-right {
    position: relative;
}

.mini-cart {
    position: absolute;
    top: calc(100% + var(--space-2));
    right: 0;
    width: 320px;
    max-width: calc(100vw - 2rem);
    z-index: 20;                        /* inside the sticky header, above the page */
    background-color: var(--color-white);
```

```
    border: 1px solid var(--color-border);
    border-top: 3px solid var(--accent); /* accent colour hsl(116, 70%, 45%) */
    box-shadow: 0 16px 30px rgba(0, 0, 0, 0.15);
    padding: var(--space-4);
}

.mini-cart[hidden] {
    display: none;
}

.mini-cart__title {
    font-family: var(--font-heading);
    text-transform: uppercase;
    font-size: var(--font-size-base);
    margin-bottom: var(--space-3);
}

.mini-cart__empty {
    font-size: var(--font-size-sm);
    color: var(--color-text-muted);
    margin-bottom: var(--space-3);
}

.mini-cart__list {
    max-height: 260px;
    overflow-y: auto;
}

/* Each row: name | price | Remove  (CSS Grid) */
.mini-cart__item {
    display: grid;
    grid-template-columns: 1fr auto;
    gap: 2px var(--space-3);
    align-items: center;
    padding: var(--space-2) 0;
    border-bottom: 1px solid var(--color-border);
    font-size: var(--font-size-sm);
}

.mini-cart__name {
    font-weight: 500;
}

.mini-cart__price {
    color: var(--color-text-muted);
    grid-column: 1;
}

.mini-cart__remove {
    grid-column: 2;
    grid-row: 1 / 3;
    font-size: var(--font-size-xs);
    text-decoration: underline;
    color: var(--color-sale);
}

.mini-cart__total {
    display: flex;
    justify-content: space-between;
    margin: var(--space-3) 0;
    font-size: var(--font-size-base);
}
```